

Relational attention for a structured semantic latent packet: toward morphology-general robot control

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251 total unique robot embodiments!

Counted by `make_figures.py` embodiments: distinct body keys built into a simulator model and simulated in at least one recorded run (validation, teacher screen, collection, training or evaluation). 150 single arms (arm $\times$ gripper; 108 procedural), 19 dual-arm rigs, 80 legged and humanoid bodies (64 procedural humanoids), the Ψ_0 G1 and the ComputerWorld pointer; 150 of them are in a training set. Breakdown: Table 8.

Abstract

We study robot control in which a planner (System 1) samples a continuous latent packet $z \in \mathbb{R}^{K \times M \times D}$, one slot per knot time and per body assembly, by rectified-flow matching, and a fast body-side controller (System 0) realizes the same packet as native joint targets from proprioception and touch alone. The packet is supervised to carry task meaning (target, contact, held object, goal) through probes. Every attention layer adds declared relational structure to its logits: the logit is $\langle q_i, k_j \rangle / \sqrt{d}$ plus a sum of factor terms (PaPE-style relative geometry, kinematic graph, contact, support and force flow, grasp, task-gated next contact, UI structure). The terms come from one declarative registry of 76 factors (75 implemented) that also drives relation-aware data generation and probe readouts, with a deploy guard that keeps simulator truth out of run-time inputs. Across arms, quadrupeds, hexapods, humanoids and a 3D UI world we record that semantic supervision raises arm competence (+0.23 [0.18, 0.28], D-134), carries a task-context halt request into legged behaviour (D-124) and lets packet edits steer a pointer (D-142), but that the latent route does not beat behaviour cloning when adapting to a new arm (D-135/D-136). A pre-registered campaign (humanoids, arm diversity, Ψ_0 , pointer, relation factors) is running; this report states its gates and no campaign result.

★ Contributions.

① **Packet interface.** System 1 $\rightarrow z[K \times M \times D] \rightarrow$ System 0, with per-assembly queries computed from public morphology (no per-ID tables); the same tensor is probed, transmitted and consumed.

② **Relational attention as a sum of declared terms.** One registry (field $\times$ operator $\times$ form $\times$ source) drives biases, kernel-compatible augmentations, gates and masks, and a deploy guard.

③ **Relation-aware data.** One privileged state interface, label and scene-part registries, factor-agnostic transforms and a progressive-composition scheduler generate the data each relation needs.

④ **A ledger.** Recorded results with their intervals, including negative ones, and a pre-registered campaign with its gates.

1 Introduction

A robot policy that is meant to move between bodies has to say what it intends in a form that survives the change of body. We take the *latent packet* as that form. A planner emits z ; a body-side controller turns the received z into joint targets using only what the body itself senses. Task meaning therefore has to live on z , and we require evidence for that from probes *and* from causal edits of the received packet.

The model is organized by a naming that follows Ψ_0 [1]: System 2 is an optional frozen vision-language model, System 1 the flow-matching planner, System 0 the packet realizer. Unlike Ψ_0 's System 0 (an off-the-shelf RL tracker), ours is a learned realizer; an RL tracker sits underneath it for legged bodies. The second ingredient is a view of attention as a sum of named relational terms. Relative geometry follows the kernel-compatible position terms of PaPE [2]; graph, contact and task terms follow the relational-inductive-bias view [5]. What is new here is bookkeeping: each term is one registry entry that also says which labels train it, which scene parts generate its data and whether it is a bias, an augmentation, a gate, a mask or only a readout.

What this report claims. (i) The design, with every factor listed (Table 2, Appendix A). (ii) Recorded results from the ledger, each with its decision number and interval (Table 4). (iii) A pre-registered campaign, described as a plan (Table 5). *Not claimed:* that relation factors help (the experiment that tests this is running), that the latent route transfers across bodies (it did not on a new arm), or any physical-robot result (simulation only).

2 Architecture

System 1. An encoder-decoder transformer trained by rectified flow [3, 4]. The encoder reads clean context tokens once per observation. Tokens come in typed banks: *morph* (one per action node, passive joint and assembly), *scene* (tracked objects with uncertainty and an explicit null token), *task* (events, role bindings, predicates, status) and *interact* (touch and receipts). Legged and humanoid bodies use one global token plus per-joint, per-limb, per-foot and terrain-cell tokens; the pointer world uses UI-widget and instruction-character tokens. Only the Ψ_0 family and an optional arm variant read an image, through System 2 (a

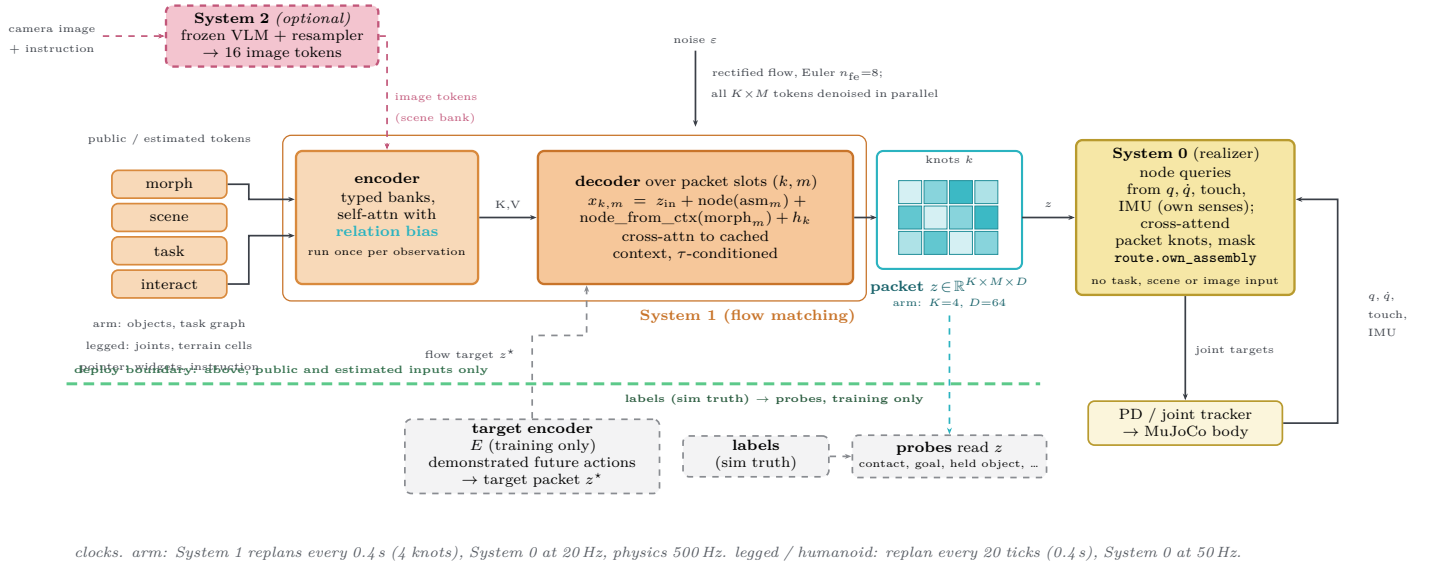


Figure 1: System overview. System 1 encodes typed context tokens (public or estimated fields only) once per observation and denoises the packet z in parallel; System 0 realizes the same z from the body’s own senses with no task, scene or image input. Dashed boxes are optional (System 2) or training-only (target encoder, labels, probes: simulator truth, never a deployed input).

Table 1: Packet shape per family ($z[K \times M \times D]$), from the code. Pointer $D=26$ with the v2 key code (T8).

family	K	knot times / horizon	M	D
arm / dual	4	0.1–0.7 s	1 / 2	64
legged / humanoid	4	0.1–0.7 s (0.8 s)	≤ 11	32
Ψ_0 (G1)	5	chunk steps 5–29 (0.6 s)	6	64
pointer	4	–	1	16 (26)

frozen VLM and a small resampler giving 16 image tokens).

The decoder’s queries are the packet slots (k, m) for knot k and assembly m ,

$$x_{k,m} = z_\tau[k, m] \text{ in-proj} + \text{node}(a_m) + \text{node_from_ctx}(c_m) + h_k, \quad (1)$$

where a_m are the assembly’s public morphology features, c_m its encoded morph token and h_k a knot embedding. The identity of a query is thus computed from public morphology, not looked up in a per-ID table, which is what lets a new body produce new queries. Each block applies self-attention over time and across assemblies, cross-attention to the cached context and an MLP, conditioned on flow time τ . Sampling integrates $\tau : 0 \rightarrow 1$ by 8 Euler steps over *all* $K \times M$ tokens in parallel; the loss is $\|v - (z - \epsilon)\|^2$ with $z_\tau = (1 - \tau)\epsilon + \tau z$. A training-only target encoder E turns demonstrated future actions into the target packet; probes read meaning out of z and supervise it.

System 0. Each control tick, one query per actuated node is built from public morphology and the *current* measured state ($q, \dot{q}$; legs add IMU, own-foot touch and an oscillator). The node queries cross-attend to the packet’s knot tokens, whose times are shifted by the elapsed phase, through the routing mask `route.own_assembly` (a node reads only its own assembly’s knots; legged bodies add the body assembly’s); then self-attention among nodes and an MLP give one nor-

malized joint target per node. System 0 receives *no* task, instruction, scene, image or System 1 state: task meaning arrives only through z , sensor feedback every tick. Clocks: the arm replans every 0.4 s and runs System 0 at 20 Hz over 500 Hz physics; legged and humanoid bodies replan every 0.4 s with a 0.8 s packet at 50 Hz.

Deploy boundary. Every token set separates *fields* (public or estimated) from *labels* (simulator truth). Each field and factor carries a provenance, *public*, *estimated* (a public estimator or the factor’s own probe) or *privileged*, and a deployable policy refuses privileged sources (`assert_deployable`). Ground truth trains the probes; at run time only public and estimated values steer attention.

3 Relational attention bias

Every attention call in the model uses one shared multi-head attention whose logit is

$$A_{ij} = \frac{q_i^\top k_j}{\sqrt{d}} + \sum_{f \in \mathcal{B}} w_{f,h} \gamma_f v_f(i, j) + \langle \phi_q(i), \phi_k(j) \rangle \quad (-\infty \text{ if masked}), \quad (2)$$

where each registered factor f contributes one term $v_f(i, j)$ in one *form*: an additive *bias* ($f \in \mathcal{B}$), a kernel-compatible *augmentation* appended to q, k (the $\langle \phi_q, \phi_k \rangle$ term, so fused attention kernels still apply), a *gate* $\gamma_f = 2\sigma(u \cdot c + b)$ that scales another term by a task, goal or embodiment summary c , or a hard *mask*. Three non-attention forms (message, embed, readout) reuse the same entries. Every learned coefficient is zero-initialized, so switching a factor on in a trained model is a no-op at step 0.

Operators. A factor is an entry over about a dozen operators, which are the only code per relation *kind*: `diff`, `sqdiff+diff` (PaPE), `rel_rot`, `align`, `order`, `same`, `sim`, `edge`,

Family	Impl.	Data gen	Attn	Readout	Planned
structure	9	0	7	0	4
geometry	5	5	5	4	3
kinematics/body	8	0	8	0	1
interaction	5	5	5	4	6
task/procedure	15	1	15	1	3
locomotion	2	2	1	2	0
UI	5	5	5	1	0
time	0	0	0	0	3
probe readouts	26	0	0	26	0
prop./aff./lang.	0	0	0	0	8
total	75	18	46	38	28

Generated by `make_figures.py` from the registries at `c18101e8`.

Table 2: Relation factors by family. Counts are computed from the registry; a factor can count in several columns. Planned = declared in the registry or catalogued (waves W2, P), not implemented. The full per-factor table is Table 6.

hop, **ancestor** (closure), **flow**, **bilinear** and **unary**. The PaPE term in its compact kernel-compatible form is

$$v_{\text{PaPE}}(i, j) = -(r_j - r_i)^\top M_i (r_j - r_i) + \tilde{b}_i^\top (r_j - r_i), \quad (3)$$

with $M_i \succeq 0$ diagonal in a learned frame and $\tilde{b}_i$ a learned direction; it splits into a *distance* part and a *direction* part (the first two panels of Fig. 2). Graph terms come from declared morphology (`kin.ancestor`, siblings, same assembly) and from the supplied task graph (actor, patient, destination edges). Pair terms that cannot be read from public inputs (contact, support, held-by, handover, next contact) are **bilinear**: $\langle Ux_i, Vx_j \rangle$ is at once the attention term and the pair probe, so the bias *is* a supervised attention subspace, and its estimate can feed a graph factor (`ix.support`→`ix.force_flow`) with source *probe*, deployable. `task.next_contact` is gated by the task so that the same scene shows several candidate edges and the bias narrows to the one the task needs.

Sites and controls. Factors attach at the encoder (`ctx>ctx`), decoder cross- and self-attention (`act>ctx`, `act>act`) and System 0 routing (`node>knot`). Uniform controls apply to any factor: **off**, **zero**, **rewired** (keys permuted, degree preserving), **shuffled**, **reversed**, and for message factors **serialized**, the same facts as hashed text (the equal-information baseline).

Probes and diagnostics. Probes are readouts in the same registry. Attention maps and probe accuracy are diagnostics: causal use is shown by editing the packet or the context and rolling out (Section 6, D-142, D-124). Fig. 2 shows factor *values* computed from real simulator states, not learned attention.

4 Relation-aware data and curriculum

Each relation may need data that decouples one invariance, so data generation is as declarative as the factors. A single privileged **StateView** (capability `privileged_truth`, implemented per backend) exposes poses, contacts, joints, cameras and UI trees for *supervision only*. Four registries

sit on it: **LABELS** (18 label functions, each with a provenance `gt`, `task` or `estimator`), **PARTS** (7 composable scene components: `table_objects`, `camera_depth`, `grasp_target`, `stack`, `terrain_steps`, `cw_viewport`, `cw_depth`, each declaring the dynamics it activates, requires and conflicts with), **TRANSFORMS** (6 factor-agnostic sample transforms: `reveal`, `surprise`, `cf_swap`, `noise`, `occlude`, `subsample`) and one composition operator that returns a scene covering a requested set of active dynamics. A factor entry names its **label** and its **gen** parts or transforms; decoupling pairs are copies of a scene that differ only in one factor’s value.

Progressive composition. One scheduler samples the generated rows. Every factor starts at composition depth $k=1$ (isolated parts); it is promoted to $k+1$ when its competence $c_f \geq \theta_f$ for two consecutive intervals; per-factor shares are $\propto s_{\min} + s_f$, clipped to [2%, 35%], with struggle $s_f = (1 - c_f) + 0.5p_f + i_f + e_f$ (plateau, interference and attributed failures); the full-task share follows a rising floor. Steering is a small grammar (boost, pin, freeze, drop, revert) logged for exact replay. *State*: implemented and plumbing-smoked; the campaign run that measures competence by depth and interference (T9) is in progress, so there is no curriculum result in this report. The `cf_swap`, `reveal` and `surprise` transforms produce the decoupling and progressive-information variants of the contact and next-contact labels.

5 Environments, bodies and tasks

All simulation is native MuJoCo [6] with robot models largely from MuJoCo Menagerie [7] (plus a batched Warp legged env and the Ψ_0 SIMPLE benchmark on Isaac Sim) behind one environment, one policy and one task interface; one harness runs any accepted policy $\times$ environment $\times$ task triple and records declined pairs with reasons. Table 3 lists the bodies and tasks and Fig. 3 shows one frame of each environment. Evaluation routes use matched seeds: R0 scripted teacher (privileged), R1 oracle packet *E*(expert chunk) (diagnostic, not deployable), R2 generated packet (deployable), plus a behaviour-cloning (BC) positive control and task-context edit suites. Splits are sealed before results.

6 Results so far

Table 4 lists every result recorded to the snapshot of 2 Oct 2026, with the setting, the comparison and the interval recorded in the cited decision. Results come from three environments (arm pick-place, legged locomotion, ComputerWorld pointer) and one third-party humanoid policy; none of them is a cross-body transfer claim. Source labels follow the repository convention: *learned latent* and *learned BC* are policies under test; *scripted teacher* and *privileged oracle* are references.

Semantic packet supervision helps within the training bodies. On four arm bodies and two training seeds the latent route with semfix succeeds in 425/480 R2 episodes against 314/480 for the capacity-matched nosem control, a difference of +0.23 [0.18, 0.28], with semfix ahead in 8/8 body $\times$ seed cells (D-134). The latent route stays be-

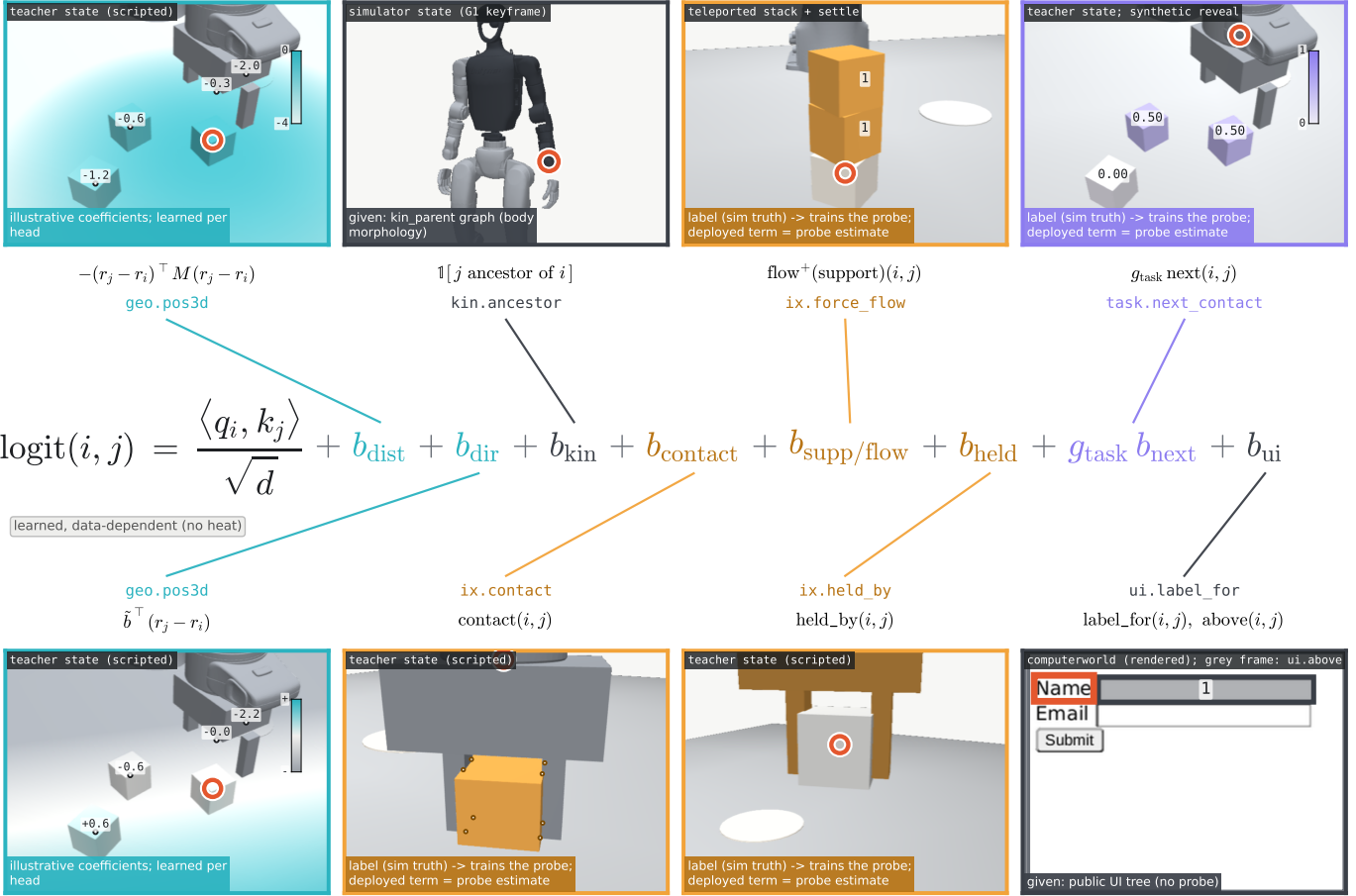


Figure 2: The attention logit, one term at a time. Each panel stretches one term of $\text{logit}(i, j)$ for a single query token (ring) over a real simulator state; frame colour is the relation family (geometry, kinematics/UI, contact/support/grasp, procedure). The content term $\langle q_i, k_j \rangle / \sqrt{d}$ is learned and data-dependent, so it has no panel. *The values shown are computed by the factor operators of `rrp.policies.relations` on real simulator states; they are not trained attention.* Panels 1–2 run the PaPE kernel with hand-set, illustrative coefficients (the model learns them per head); panels 3 and 8 are public, given structure (G1 kinematic tree; computerworld widget tree). For probe-sourced pair terms (panels 4–7) the colour shows the simulator-truth label that trains the probe; the deployed term is the probe’s estimate, which is not shown. Scenes: scripted-teacher Panda pick-and-place states (1, 2, 4, 6, 7), a teleported three-cube stack (5), a G1 keyframe (3), a rendered desktop UI (8). Terms whose form is *aug* (`geo.pos3d`, `ix.contact`, `ix.held_by`, `task.next_contact`) enter the logit as $\langle \phi_q(i), \phi_k(j) \rangle$ in Eq. (2); the panel shows that value. Planned relation factors (e.g. `time.same_track`) are omitted.

low the BC control on panda (135/180 against 58/60) and the binding (rebind) effect is seed-dependent for both variants. An earlier “semantic supervision hurts” result was a confound: the unbounded probe loss shrank System-0 updates to ≈ 0.002 – 0.004 against 0.70 – 0.92 for nosem; with the bounded loss the three variants read 81/90, 83/90 and 21/90 (bounded, nosem, original; D-085, D-087). Comparisons made before that fix are not used here.

The packet is causally used, in two environments. In legged locomotion, adding a “halt” request to System 1’s task context (System 0 has no task input, so the request can reach the body only through z) changes forward progress over 2–5 s by -0.40 m [$-0.50, -0.30$] on `anymal_c` and -0.66 m [$-0.72, -0.59$] on `go2` for `semfix` relative to `nosem`, ordered in 3/3 seeds, while an irrelevant context change moves the body by at most 0.02 m (D-124). In ComputerWorld a probe-guided edit reduces the distance to the original target from 163 px to 41 and 22 px for `semfix`; a random edit of equal norm gives

102–105 px, and `nosem` edits behave like random ones (D-142). The edits are partial (the probe reads the new target in 8–10 of 25 episodes) and task success is unchanged. These are rollouts after interventions, not probe accuracies.

Zero-shot transfer to a new arm is not supported. On sealed `xarm7` targets, latent and BC policies both score 0/200 zero-shot. With target demonstrations BC fine-tuning reaches 69–192 of 200 depending on target, whereas flow fine-tuning gives 0 and a System-0 refit at most 38 (D-135). Joint flow and realizer adaptation, added *after* D-135 and therefore post-hoc, beats the refit in 12/12 cells (595 against 130 of 1,200 for `semfix`) but stays below BC fine-tuning in 12/12 (595 against 842), with every Newcombe upper bound below zero (D-136). On the known arm with a new gripper (`panda_tf3`) `semfix` reaches 154/200 zero-shot, `nosem` 32 and BC 199 (D-135). A dev-seed diagnosis (D-135 addendum) attributes the D-135 failure to the protocol adapting one module at a time and not to an inability of the latent route to

Table 3: Environments, bodies, tasks, the relation families whose factors resolve in each env’s net family (implemented factors: total, net-side + probe), and the status of each track.

env	bodies	tasks	relation family	status
arm mujoco/arm	panda, sawyer, ur10e, ur5e, vx300s, parm5/6/7 variants (5), 24 procedural pa2s* (pg2 / tf3 grippers); sealed targets panda_tf3, xarm7	pick_place, reach_pose	arm 42 (33+9)	teacher + learned; transfer split primary_v1 / armdiv
dual mujoco/dual	two mounted arms (pairs of the arm pool) or one dual body	support_insert, handover, assign_left, assign_right, pivot_against_surface	dual 43 (34+9)	parked, D-146: eval of external checkpoints; scripted teachers only
quad mujoco/legged	go2, anymal_c, a1, hexapod6, pquad4, sprawl4, sprawl8, hexapod6_long	waypoint_contact, foothold_steps	legged 21 (16+5)	teacher + frozen tracker
humanoid mujoco/legged	t1, g1, h1, op3, apollo, adam_lite, talos + phum procedural	9 h_* tasks (h_steps, h_gap, h_walk, ...); held out: h_steps_carry, h_gap_cart	humanoid 21 (16+5)	teacher + frozen tracker; campaign in progress
tracker warp/legged	same legged and humanoid bodies, GPU-batched	locomotion, h_steps, h_gap	legged / humanoid (as above)	tracker training only
SIMPLE simple	g1_simple (G1, 36-d command)	6 G1Wholebody* tasks	psi0 19 (10+9)	Ψ_0 SIMPLE on Isaac Sim; path-traced renderer caveat; step-1 reproduced on 3 of 6 tasks
CWorld computerworld	cw_pointer (cursor, keys)	calc_sum, open_type, drag_window, fill_form	pointer 10 (7+3)	scripted teacher; procedural strings v3

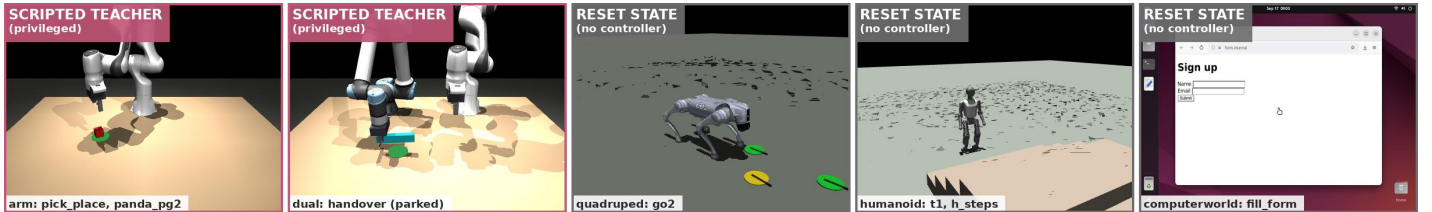


Figure 3: The five environments. One rendered frame each. The arm and dual-arm frames are mid-episode states of the scripted teacher (privileged source); the quadruped, humanoid and computerworld frames are reset states with no controller running, so they show the scene, not a rollout.

adapt; even so, on dev seeds the latent route adapts about half as fast per update as BC and never beats it on a new body in the data so far.

Pointer control: close to BC in-distribution, no copy on unseen strings. On the sealed `sealed_id` set ($n=400$, one training seed, sealed once) BC solves 398 (0.995 [0.982, 0.999]), semfix 372 (0.930 [0.901, 0.951]) and nosem 368 (0.920 [0.889, 0.943]); the scripted System-0 route (`eng`) solves 185 (D-142). The semfix–nosem difference is not distinguishable ($p=0.68$, McNemar); BC is ahead of semfix ($p=2.2 \times 10^{-7}$). On unseen words and names every learned method scores 0/50 on typing and 0/50 on form filling; unseen calculator pairs reach 45–50/50. The learned realizers do not copy characters from the instruction; a copy head is the pre-registered test (Sec. 7).

Structure on a pretrained humanoid policy: a failed diagnosis, not a negative result. On the released Ψ_0 checkpoint, 20 closed-loop episodes give 20/20 [0.84, 1.00] released, 19/20 [0.76, 0.99] direct fine-tuning and 0/20 [0.00, 0.16] with the structured head (D-141). The offline diagnosis traced the 0/20 to an integration fault: System 0 barely read the packet and acted as a state-only policy, driven by a commanded-torso state that never occurs in training; so it is

not evidence against structure. The pre-registered packet-use gate that was added afterwards passed (gap 0.0782; D-146, D-147), and the closed-loop comparison is queued (Sec. 7).

7 Pre-registered campaign

The remaining claims are decided by a ten-node campaign (T0–T9; D-147, `research/readiness.md` section B) whose tests and reading rules were written before the corresponding runs. Table 5 gives each node’s state at the snapshot. GPU work is admitted by a resource broker (at most two concurrent GPU jobs per machine, humanoid first); a failed gate is recorded and not retried with other settings. Sealed splits are hash-pinned and each cell runs once; infeasible or failed cells are reported as such.

Humanoid (T0–T5). Question: with pinned, gated trackers, does the legged relation preset change dev transfer relative to the same recipe without it (`preset:legged` against `legged-none`, at least two seeds, matched data and updates)? A tracker is installed only if it passes the D-112 gate, or the labelled D-147 exception (joint margin ≥ -0.06 , peak force ≤ 4.2 body weights, cost of transport ≤ 2.5 , task evaluation ≥ 0.9). One bounded fix per failed gate is al-

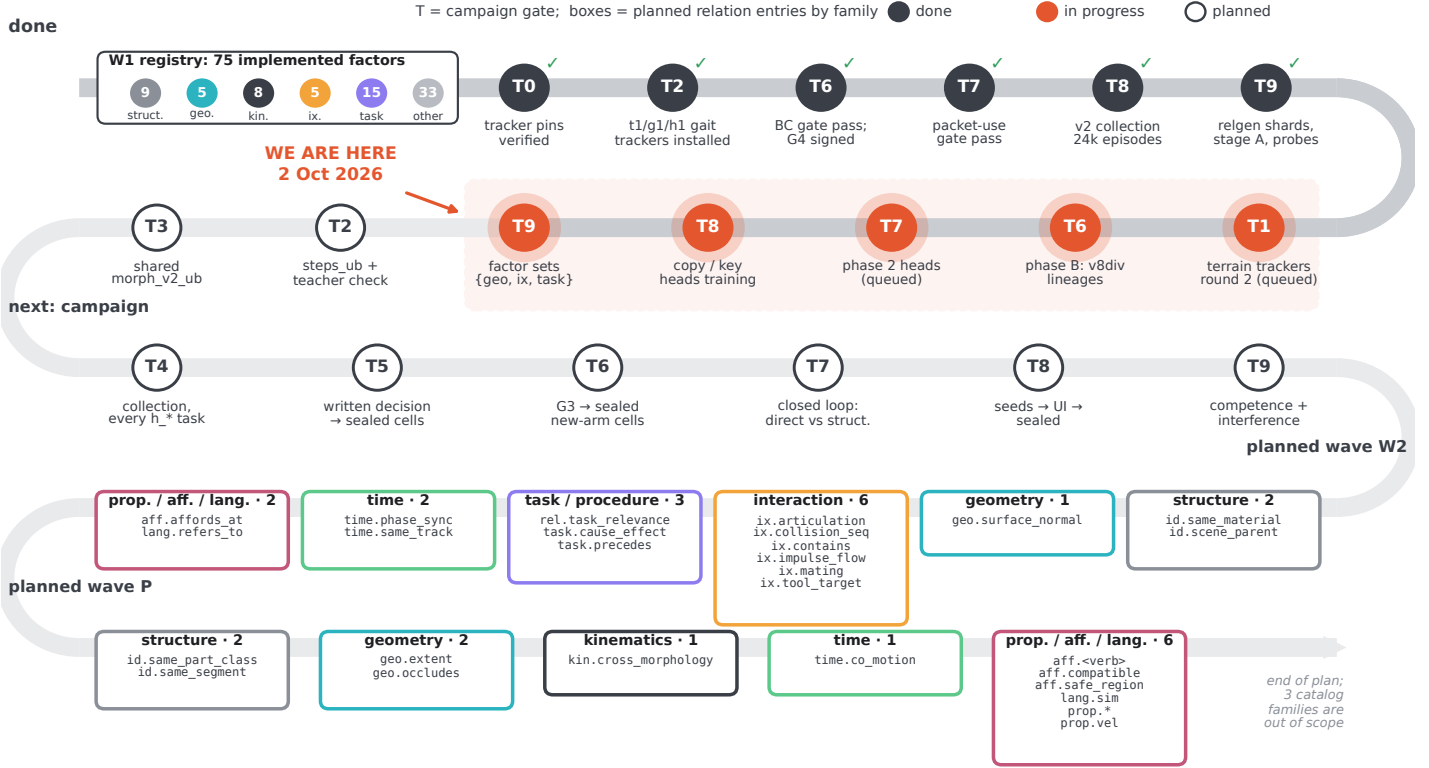


Figure 4: Roadmap as of 2 Oct 2026. The path runs from completed work (top) to the end of the current plan (bottom). Row 1: the 75 implemented relation factors by family and the campaign gates already passed (T0 pins, T2 gait trackers, T6 BC gate and G4 signature, T7 packet-use gate, T8 collection, T9 shards and stage A). Row 2: the current front (highlighted; T1 round 2, T6 phase B lineages, T7 phase 2, T8 copy and key heads, T9 factor sets), then the next humanoid gates. Row 3: the remaining campaign gates, each ending in sealed or pre-registered readouts (Table 5). Rows 4–5: every planned entry of Table 7, clustered by its family, wave W2 before wave P; they come after T9 because T9 is the first test of whether factor sets help. Colours follow the families of Fig. 2 (geometry cyan, kinematics graphite, interaction amber, procedure violet; structure, time and property/affordance/language get neutral extra hues). Filled = done, highlighted = in progress, hollow = planned. Generated by `make_figures.py roadmap` from the relation registry and catalog.

lowed. Terrain-scan trackers failed round 1 (T1), so terrain tasks are currently *unsupported*; a task with teacher success below 0.8 is held back (T4). The test is the dev transfer tables followed by a written decision before any sealed cell (T5); no humanoid transfer claim is made until then.

Arm diversity (T6). H1: a latent route trained on 65 distinct arm kinematics transfers to new arms better than the v6 13-body lineages, zero-shot and at 5, 20 and 100 target demonstrations, and closes the gap to BC fine-tuning. H2: kinematic static features (`kinfeat`) help further. Success is $k/100$ per cell with Wilson intervals; three Newcombe contrasts: (a) v8div – v6 latent per primary target, (b) v8div latent – v8div BC fine-tuning at equal budget, (c) kinfeat – no-kinfeat. A contrast counts only if its interval excludes 0; tiers are not pooled; zero-shot is reported apart from adaptation. Gate G3 requires v8div semfix $\geq$ v6 semfix -0.05 on the original bodies and a BC expert ≥ 0.85 on the new arms (BC half passed). A null or negative H1 is recorded as `failed_hypothesis`; targets and budgets are not changed.

Packet use in Ψ_0 (T7). Question: does a structured head that reads the packet change closed-loop pick-place relative to direct fine-tuning? The gate requires

$\text{err } R(\bar{z}) - \text{err } R(E(a)) \geq 0.05$; a failure ends the node as `failed_hypothesis`. It passed (0.0782). No numerical threshold is recorded for the closed-loop comparison (20 episodes; D-141: released 20/20, direct 19/20), so it is reported with intervals and not as pass or fail.

ComputerWorld pointer (T8). Dev seeds, 50 per task, three training seeds; a difference counts only if it has the same sign in at least two of three paired seeds and exceeds the larger within-arm seed range. P-SEEDS: is the D-142 ordering stable? P-UI: do ui.* factors help (mean gain ≥ 4 points on ≥ 2 tasks, no task losing more than the seed range)? P-COPY: does a copy head fix unseen typing ($\geq 10/50$ on both `open_type` and `fill_form` of `sealed_heldout`)? P-KEY: does a discrete key code help (dev gain above the seed range)? Sealed sets are evaluated once per arm and seed after the dev table is written. The scripted route cannot copy and is excluded from P-COPY.

Relation factors (T9). Question: do geometry, interaction and task factor sets added at stage A give probes with measurable competence, and do they interfere? The recorded tests are per-factor competence by curriculum depth and an interference table against the base set (two

seeds). **No numerical success threshold is recorded for T9**; it can establish only that the diagnostics work, not downstream performance. Probes and attention maps are diagnostics; causal use is shown by edits and rollouts (Sec. 6).

8 Limitations

The evidence is narrow. (i) *No cross-body transfer claim*. Zero-shot transfer to new arms is 0/200 for every method, the latent route has not beaten BC on any new body, and the humanoid transfer sealed cells have not been run. (ii) *Seeds*. Most ComputerWorld numbers are one training seed (D-142); the legged result has three seeds per body (D-124), with exact seed-ordering $p=0.05$. Intervals cover evaluation episodes, not training-seed variance. (iii) *Post-hoc analysis*. The joint-adaptation method of D-136 was added after the sealed results of D-135 on the same scenes; it is labelled and not a pre-registered test. (iv) *Partial edits*. Packet edits move behaviour part of the way (the probe reads the edit in 8–10 of 25 cases) and are demonstrated on calculator and form tasks only. (v) *Unseen strings*. All learned pointer methods fail on unseen typing and form text (0/100); this is a negative result about the current realizers. (vi) *Pretrained humanoid*. The structured arm’s 0/20 was an integration fault, fixed only offline so far; the closed-loop comparison has not run. (vii) *Humanoid trackers*. Terrain trackers failed their gates, and the three installed gait trackers (t1, g1, h1) are labelled exceptions to the joint-margin rule. (viii) *Simulation*. All results are simulated, with scripted or privileged teachers; sim-truth labels train probes and are not deployable inputs. No physical robot was commanded. (ix) *Snapshot*. The campaign table is a snapshot; running nodes may finish with results that are not shown here.

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A Relation tables (generated)

Both tables are written by `make_figures.py table_relations` from the live registries and the candidate catalog; nothing about an implemented factor is typed by hand.

B Embodiment count (generated)

Table 8 is written by `make_figures.py embodiments`, which reads the teacher-validation records, the arm admission screens, the dual-arm teacher validation, the legged body catalog, the tracker training metadata (host store and the host copy of the peer archive) and the frozen splits, and keeps the per-class key lists in `figures/embodiments.json`.

Embodiment class	Sim.	Trained	Evidence
single arms: menagerie + fixture arms, each with each gripper module (pg2, tf3)	42	17	teacher validation (20 seeds) + armdiv admission screens
procedural arms <code>arm_gen_v2</code> (<code>pa2s<seed></code> , train and sealed seeds) $\times$ gripper	108	48	armdiv admission screens (built keys only)
dual-arm rigs (unordered arm pairs) + ALOHA	19	6	dual teacher validation, assign split
procedural legged (quad / hexapod / octopod)	5	5	legged catalog (physics validated)
menagerie quadrupeds and bipeds	4	2	legged catalog (physics validated)
menagerie humanoids	7	6	legged catalog + tracker training metas
procedural humanoids <code>phum_<seed></code>	64	64	shared-tracker training metas
G1 + Dex3 through the Ψ_0 / SIMPLE interface (<code>g1_simple</code>)	1	1	D-141 closed-loop evals
ComputerWorld pointer (<code>cw_pointer</code>)	1	1	D-142 evals
total	251	150	

Generated by `make_figures.py embodiments` at 908a0430; keys in `figures/embodiments.json`.

Table 8: Embodiments counted for the banner. Sim. = distinct body keys built into a simulator model and simulated in at least one recorded run (validation, teacher screen, collection, training or evaluation); Trained = of those, in a frozen training pool or a tracker’s training set. Arm keys are arm $\times$ gripper module; procedural arms are 54 built `arm_gen_v2` seeds; generator seeds whose build failed, dual left/right swaps, and declared but not yet run bodies (humanoid sealed targets, `phum` sealed seeds) are not counted.

Table 4: Results so far (recorded results only; interim numbers of running campaign nodes are not listed). “Latent” = learned System 1 $\rightarrow$ packet $\rightarrow$ learned System 0, deployable route (R2); “BC” = learned behaviour-cloning control on the same demonstrations; “teacher”, “oracle” = scripted / privileged references, never policies under test. Intervals are the 95% intervals recorded in the cited decision (Wilson for rates, Newcombe for rate differences); “counts” = no interval recorded. semfix = semantic packet supervision with the bounded probe NLL (D-085); nosem = capacity-matched control without it.

Question	Setting (env, bodies, seeds, n)	Comparison	Result [95% CI as recorded]	Decision
Does semantic packet supervision help the arm latent route?	Arm pick-place, 4 bodies $\times$ 2 training seeds, 240 R2 episodes per lineage (480 pooled per variant); v6 data, teacher v2, grasp v2.1	latent semfix vs latent nosem; BC reference	semfix 425/480 (0.89) vs nosem 314/480 (0.65): +0.23 [0.18, 0.28]; semfix wins 8/8 body $\times$ seed cells. Binding (rebind) is seed-dependent for both. Latent is below BC on panda (semfix 135/180; BC 58/60; BC 60/60 on parm6).	D-134
Does the latent route transfer <i>zero-shot</i> to a new arm?	Sealed targets xarm7_pg2, xarm7_tf3 (new arm), panda_tf3 (new gripper, known arm); 2 seeds $\times$ 100 sealed scenes, k/200 pooled	latent semfix / nosem vs BC; zero-shot, then flow SFT, system-0 refit, BC SFT at 5/20/100 target demos	xarm7: zero-shot 0/200 for latent and BC; BC SFT 69–140 (pg2), 143–192 (tf3); latent refit $\leq$ 38, flow SFT 0. panda_tf3: semfix zero-shot 154, nosem 32, BC 199. counts. Not supported.	D-135
Does joint flow + realizer adaptation close the gap? (<i>post-hoc</i>)	Same sealed xarm7 scenes; 2 seeds; 6 target $\times$ budget cells per variant (1,200 episodes); method added <i>after</i> D-135	joint adaptation (update-matched 150/300/600) vs refit-only vs BC SFT	Beats refit in 12/12 cells (semfix 595 vs 130 of 1,200); below BC SFT in 12/12 cells (semfix 595 vs 842; gap -0.07 to -0.34 ; nosem -0.30 to -0.80 ; every Newcombe upper bound < 0). semfix 595 vs nosem 151. Latent never beats BC.	D-136
Does the packet carry a task-context “halt” into legged behaviour?	mujoco/legged, contact v2, anymal_c and go2, 3 training seeds each; R2 success $n=90$ per variant; forward-progress change over 2–5 s	latent semfix – nosem, pooled; halt request added to the task context	anymal_c -0.40 m [-0.50 , -0.30]; go2 -0.66 [-0.72 , -0.59]; 3/3 seeds ordered (exact $p=0.05$); irrelevant-context control ≤ 0.02 m; success comparable (anymal_c 78 vs 83/90, go2 87 vs 87). t1 under sourced limits: untestable (BC 0/30).	D-124
Was “semantic supervision hurts” real? (confound found and fixed)	t1 humanoid (inflated-torque tracker), R2, 3 training seeds, $n=90$	bounded-NLL sem vs nosem vs original sem	81/90 vs 83/90 vs 21/90. The unbounded probe NLL scaled System-0 updates by ≈ 0.002 – 0.004 (nosem 0.70–0.92); earlier sem-vs-nosem comparisons were confounded. counts.	D-085, D-087
Pointer on ComputerWorld: how does the latent route compare?	4 tasks, sealed_id $n=400$ (100/task), 1 training seed, sealed once; teacher 400/400, oracle $\rightarrow$ scripted System 0 400/400 (references)	latent semfix vs nosem vs BC vs scripted-System-0 route (eng)	BC 398 (0.995 [0.982, 0.999]); semfix 372 (0.930 [0.901, 0.951]); nosem 368 (0.920 [0.889, 0.943]); eng 185 (0.463). McNemar: BC vs semfix $p=2.2e-7$; semfix vs nosem $p=0.68$.	D-142
...unseen words and names	sealed_heldout $n=150$ (50 calc / type / form), 1 seed	all learned methods	Typing and form-filling 0/50 each (0/100 unseen words and names) for every learned method; unseen calculator pairs 45–50/50. No method copies characters from the instruction.	D-142
...causal use of the packet	Probe-guided packet edit, 25 dev episodes each on calc and form, 1 seed	semfix vs nosem vs random edit of equal norm	Approach to the original target: 163px (no edit) / 102–105 (random) $\rightarrow$ 41 / 22 (semfix); nosem edits $\approx$ random (128–141 vs 137–144px). Edits partial (probe reads the new target in 8–10/25). Task success unchanged.	D-142
Does adding structure to a pretrained humanoid VLA work?	Ψ_0 TabletopGraspMP L0 (SIMPLE / Isaac Sim 5.1, path-traced), 10 eval configs $\times$ 2 repeats, $n=20$ per arm	released Ψ_0 (third-party) vs direct fine-tune vs structured	Released 20/20 [0.84, 1.00]; direct 19/20 [0.76, 0.99]; structured 0/20 [0.00, 0.16], diagnosed as an integration bug (System 0 ignores the packet); not evidence against structure.	D-141
...after the fix: packet-use gate	T7 phase 1, held-out episodes; gate = $\text{err}(R(\bar{z})) - \text{err}(R(E(a))) \geq \text{margin } 0.05$	stage A with state dropout and permuted-packet hinge	PASS: gap 0.0782 (0.0843 vs 0.0060). Closed-loop structured evaluations are queued, not results.	D-146, D-147

Table 5: Campaign T0–T9 (D-147; gates from `research/readiness.md` section B and the D-147 addenda), state as of 2 Oct 2026 14:00 PDT. States use the repository’s progress vocabulary only. A recorded gate outcome is listed when one exists; counts of nodes that are still `running` are deliberately omitted. “L” = body leg length (step heights are given in units of L).

	Track and run	Pre-registered gate	State	Recorded gate outcome
T0	humanoid: pinned trackers and warm starts	<code>t1:contact_v2</code> and warm-start checkpoints present with the recorded sha256	<code>verified</code>	Pins match; its D-112 report reproduces D-114 (no new run).
T1	humanoid: terrain-scan trackers, <code>h_steps</code> (t1, g1, h1) and <code>h_gap</code> (t1, h1)	D-112 gate; task eval ≥ 0.9 per height/level; labelled exception only for joint margin, peak force, CoT within caps (D-147); one bounded fix per failed gate	<code>failed_hypothesis</code> (round 1); round 2 <code>running</code> (dispatcher queued for a GPU)	steps t1 2/20 at 0.10L, 0/20 above; g1 0/20 at every height; h1 10/20 at 0.10 and 0.15L, 0/20 above; gap t1 0/20; gap h1 20/20 at level 1.0 but stand-trial falls and force above cap. None installable.
T2	humanoid: wholebody <code>*_ub</code> gait trackers (t1, g1, h1), then <code>steps_ub</code> trackers	Lab gate; teacher check <code>h_carry</code> $\geq 10/12$ both sides, <code>h_steps_carry</code> $\geq 8/10$, else <code>blocked_external</code> and excluded	<code>gait trackers completed</code> ; <code>steps_ub planned</code>	t1, g1, h1 <code>ub_v1</code> gait trackers installed under the labelled D-147 exception (lab gate pass, no falls; only the joint margin fails: 0.0183 / 0.0191 / 0.0160 against 0.02). Teacher check not yet run.
T3	humanoid: shared <code>morph_v2_ub</code> tracker	Gate on every source body	<code>planned</code>	—
T4	humanoid: collection for every <code>h_*</code> task	Teacher success ≥ 0.8 per task, else the task is held back with a note	<code>planned</code>	—
T5	humanoid: lineages, <code>preset:legged</code> vs <code>legged-none</code> , ≥ 2 seeds, dev transfer tables	Dev tables, then a <i>written decision</i> before any sealed cell (D-138 methods, matched data and updates)	<code>planned</code>	No sealed humanoid cell has run.
T6	armdiv: BC 1702, kinfeat BC 1701; G4 signature; v8div lineages $\rightarrow$ G3 $\rightarrow$ sealed	BC expert ≥ 0.85 on new training arms; v8div semfix $\geq$ v6 semfix -0.05 on original bodies; G4 protocol signed before any lineage; each sealed cell once	<code>running</code> (phase B: lineage stage A)	BC gate PASS : 117/120 new arms for both BC experts, 59/60 and 60/60 on v6 bodies, 60/60 held-out. G4 signed 2 Oct. Latent half of G3 and sealed cells: not yet.
T7	Ψ_0 : labels $\rightarrow$ stage A $\rightarrow$ packet-use gate $\rightarrow$ structured head $\rightarrow$ closed loop	Gate gap $\geq$ margin 0.05, else <code>failed_hypothesis</code> and nothing trains after it	phase 1 <code>completed</code> ; phase 2 <code>running</code>	Gate PASS : gap 0.0782 (err $R(\bar{z})$ 0.0843 vs err $R(E)$ 0.0060). Closed-loop direct vs structured: not yet.
T8	pointer v2: collection $\rightarrow$ copy / key heads $\rightarrow$ seeds $\rightarrow$ UI factors, dev tables	P-KEY, P-COPY, P-SEEDS, P-UI rules (Sec. 7); sealed only after a written decision	collection <code>completed</code> ; copy <code>running</code> ; seeds, UI <code>planned</code>	Collection: 4 $\times$ 6,000 teacher episodes, 0 teacher failures. Oracle packets $\rightarrow$ scripted System 0: 50/50 on each task (privileged reference, not a policy result).
T9	relation factors: relgen shards $\rightarrow$ shared stage A $\rightarrow$ factor sets {geo, ix, task} vs base, seeds 1, 2	Per-factor competence by depth; interference table	shards, stage A, probes <code>completed</code> ; factor-set training <code>running</code>	None yet.

Factor	Operator	Form	Source(s)	Data gen (parts / transforms)	Attn	Read	Nets	Data envs
STRUCTURE / EDGES (9 implemented)								
edge.node_in_assembly	edge	bias	given	N	Y	N	arm,dual,leg,hum	–
edge.same_assembly	edge	bias	given	N	Y	N	leg,hum,psi0	–
edge.same_node	edge	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
id.same_assembly	same	aug	given	N	Y	N	arm,dual,leg,hum,psi0	–
id.same_body	same	aug	given	N	Y	N	arm,dual	–
id.slot_handle	inert	embed	given	N	N	N	arm,dual	–
msg.incidence	inert	message	given	N	N	N	arm,dual	–
route.assembly_reads	same	mask	given	N	Y	N	arm,dual,leg,hum,psi0	–
route.own_assembly	same	mask	given	N	Y	N	arm,dual,leg,hum,psi0	–
GEOMETRY (5 implemented)								
geo.above	order	bias	given, gt [†]	Y: table_objects	Y	N	arm,dual,ptr	arm,dual
geo.depth3d	sqdiff+diff	aug	probe, given, gt [†]	Y: camera_depth	Y	Y	arm,dual	arm,dual
geo.normal_align	align	aug	given, probe, gt [†]	Y: contact_normal	Y	Y	arm,dual	arm,dual, mj-leg,wp-leg, simple
geo.orient	rel_rot	aug	given, probe, gt [†]	Y: table_objects	Y	Y	arm,dual	arm,dual
geo.pos3d	sqdiff+diff	aug	given, probe, gt [†]	Y: table_objects	Y	Y	arm,dual,ptr	arm,dual
KINEMATICS / BODY (8 implemented)								
edge.{foot_of, limb_adjacent, over_cell} (3)	edge	bias	given	N	Y	N	leg,hum	–
edge.{kin_child, mirror} (2)	edge	bias	given	N	Y	N	leg,hum,psi0	–
edge.kin_parent	edge	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
kin.ancestor	ancestor	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
kin.sibling	hop	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
PHYSICAL INTERACTION (5 implemented)								
ix.contact	bilinear	aug	probe, gt [†]	Y: grasp_target	Y	Y	arm,dual	arm,dual
ix.force_flow	flow	bias	probe, gt [†]	Y: stack	Y	N	arm,dual	arm,dual
ix.handover	bilinear	aug	probe, gt [†]	Y: grasp_target	Y	Y	dual	arm,dual
ix.held_by	bilinear	aug	probe, gt [†]	Y: grasp_target	Y	Y	arm,dual	arm,dual
ix.support	bilinear	aug	probe, gt [†]	Y: stack	Y	Y	arm,dual	arm,dual
TASK / PROCEDURE (15 implemented)								
edge.{actor_of, consumed_by, destination_of, enables, maintained, node_actor_of, output_to, patient_of, pred_arg, produced, role_in_event, role_points_to, support_of, target_of} (14)	edge	bias	given	N	Y	N	arm,dual	–
task.next_contact	bilinear	aug	probe, gt [†]	Y: reveal, surprise	Y	Y	arm,dual	arm,dual, mj-leg,wp-leg, simple
LOCOMOTION (2 implemented)								
leg.com_support	inert	readout	given	Y: terrain_steps	N	Y	leg,hum	mj-leg,wp-leg
leg.foothold	bilinear	aug	probe, gt [†]	Y: terrain_steps	Y	Y	leg,hum	mj-leg,wp-leg
UI (5 implemented)								
ui.above	order	bias	given	Y: cw_viewport, cw_depth	Y	N	ptr	cw
ui.drag_to	bilinear	aug	probe, gt [†]	Y: cw_viewport, cw_depth	Y	Y	ptr	none [‡]
ui.focus_next	edge	bias	given, gt [†]	Y: cw_viewport, cw_depth	Y	N	ptr	cw
ui.label_for	edge	bias	given, gt [†]	Y: cw_viewport, cw_depth, reveal, surprise	Y	N	ptr	cw
ui.same_window	same	bias	given	Y: cw_viewport, cw_depth	Y	N	ptr	cw
PROBE READOUTS (26 implemented)								
probe.arm.{acting_on, focused_on, goal_effect, held_by, looking_at, observed_effect, rel_pos, subtask, visible} (9)	inert	readout	given	N	N	Y	arm,dual	–
probe.legged.{contact, disp, fall, goal, subtask} (5)	inert	readout	given	N	N	Y	leg,hum	–
probe.pointer.{phase, rel, slot} (3)	inert	readout	given	N	N	Y	ptr	–
probe.psi0.{active_hand, base_cmd, base_disp, contact, grasp_face, grasp_pt, hand_dist, lift, target_pos} (9)	inert	readout	given	N	N	Y	psi0	–

Generated by make_figures.py from the registries at c18101e8.

Table 6: Every implemented relation factor. 75 factors: 18 generate or label data, 46 act on attention logits or masks, 38 supervise a readout (several per factor possible). Rows with identical classification are merged (**edge.***, **probe.<net>.***; count in parentheses). gt[†] = gt (training only): simulator truth, never a deployable input. Data gen: a data generator is declared or a relgen label exists (scene parts upright, transforms italic); probe targets are task-pipeline packet labels, hence N. Attn: form ∈ {bias, aug, gate, mask}. Read: a readout is declared. Nets: families whose **resolve(family=...)** accepts the factor. Data envs: label caps and scene part available; [‡] no registered environment provides the caps yet.

Intended entry	Status	Catalog family	Operator $\times$ form	Needs (label caps; scene part)	Envs
STRUCTURE / EDGES (4 planned)					
<i>id.same_material</i>	planned (W2)	same-X symmetry / lattice; equivalence segmentation hierarchy	same $\times$ aug	materials ; needs material ids	all
<i>id.same_part_class</i>	planned (P)		same + edge	–	simple, arm
<i>id.same_segment</i>	planned (P)		same	needs image-patch tokens with masks	arm
<i>id.scene_parent</i>	planned (W2)		ancestor + hop	–	arm, legged, simple
GEOMETRY (3 planned)					
<i>geo.extent</i>	planned (P)	shape and scale	unary + sim	extent	arm, dual
<i>geo.occludes</i>	planned (P)	visibility	unary + edge	–	arm, dual
<i>geo.surface_normal</i>	planned (W2)	surface directions	align $\times$ aug	surface_normal	arm, dual
KINEMATICS / BODY (1 planned)					
<i>kin.cross_morphology</i>	planned (P)	mirror / morphology correspondence	same + edge	–	simple
PHYSICAL INTERACTION (6 planned)					
<i>ix.articulation</i>	planned (W2)	attachment / articulation	bilinear + align	articulation_pairs ; needs a drawer / door part	arm
<i>ix.collision_seq</i>	planned (W2)	collisions over time	flow	–	arm, dual, legged
<i>ix.contains</i>	planned (W2)	region topology; containment of content	bilinear + edge	contains_pairs ; needs bin part; bin part	arm, dual, cw
<i>ix.impulse_flow</i>	planned (W2)	force transfer	flow $\times$ bias	support_pairs , contact_forces	arm, dual
<i>ix.mating</i>	planned (W2)	mating / insertion	bilinear + align	mating_pairs ; support_insert task exists	arm
<i>ix.tool_target</i>	planned (W2)	tool use	bilinear $\times$ gate	tool_pairs ; needs a tool part	arm
TASK / PROCEDURE (3 planned)					
<i>rel.task_relevance</i>	planned (W2)	relevance / salience	unary $\times$ gate, readout	–	arm, dual, cw
<i>task.cause_effect</i>	planned (W2)	causality	edge + flow	labels from task receipts + interventions	arm, dual
<i>task.precedes</i>	planned (W2)	prerequisites	ancestor	–	arm, dual
TIME (3 planned)					
<i>time.co_motion</i>	planned (P)	motion	sim + diff	velocity labels exist	arm, dual, legged
<i>time.phase_sync</i>	planned (W2)	periodicity / phase	rel_rot + align	legged gait legs	legged
<i>time.same_track</i>	planned (registry)	correspondence (declared)	same $\times$ aug	–	–
PROPERTIES / AFFORDANCE / LANGUAGE (8 planned)					
<i>aff.<verb></i>	planned (P)	affordance scores	unary $\times$ readout	labels need per-object scripted affordance tables	arm, dual
<i>aff.affords_at</i>	planned (W2)	affordance pairs	bilinear $\times$ gate	–	arm, cw
<i>aff.compatible</i>	planned (P)	compatibility	bilinear + sim $\times$ gate	–	–
<i>aff.safe_region</i>	planned (P)	safety regions	unary	–	arm
<i>lang.refers_to</i>	planned (W2)	text grounding	bilinear	needs instruction tokens in the arm ctx	–
<i>lang.sim</i>	planned (P)	semantic similarity	sim	–	–
<i>prop.*</i>	planned (P)	physical properties	unary + sim	inertia, materials ; labels exist in MuJoCo:	arm, dual
<i>prop.vel</i>	planned (P)	fields and flows	unary	inertia, materials	–
OUT OF SCOPE (3 families; no physics or sensing for them in our environments): fields and flows; fluids, thermal, electrical, magnetic, optical, chemical; mechanics.					

Generated by `make_figures.py` from `research/relations_catalog.md` and the registries at `c18101e8`.

Table 7: Planned and out-of-scope relations. 28 planned entries (1 declared in the registry with status planned, 27 from the candidate catalog, waves W2 and P); 3 out-of-scope families. Italic names are proposed (the catalog names no entry). None is implemented, so no result in this paper depends on them. Meta properties of relations (catalog section K and 3 rows tagged M, e.g. joint states, costs) are not factors and are excluded; 4 epistemic mechanisms (confidence scaling, soft edges, the reveal / surprise / noise / occlude transforms) are implemented but are not attention factors.